ECE 143
3/13/26

Group 20 Final Project Presentation

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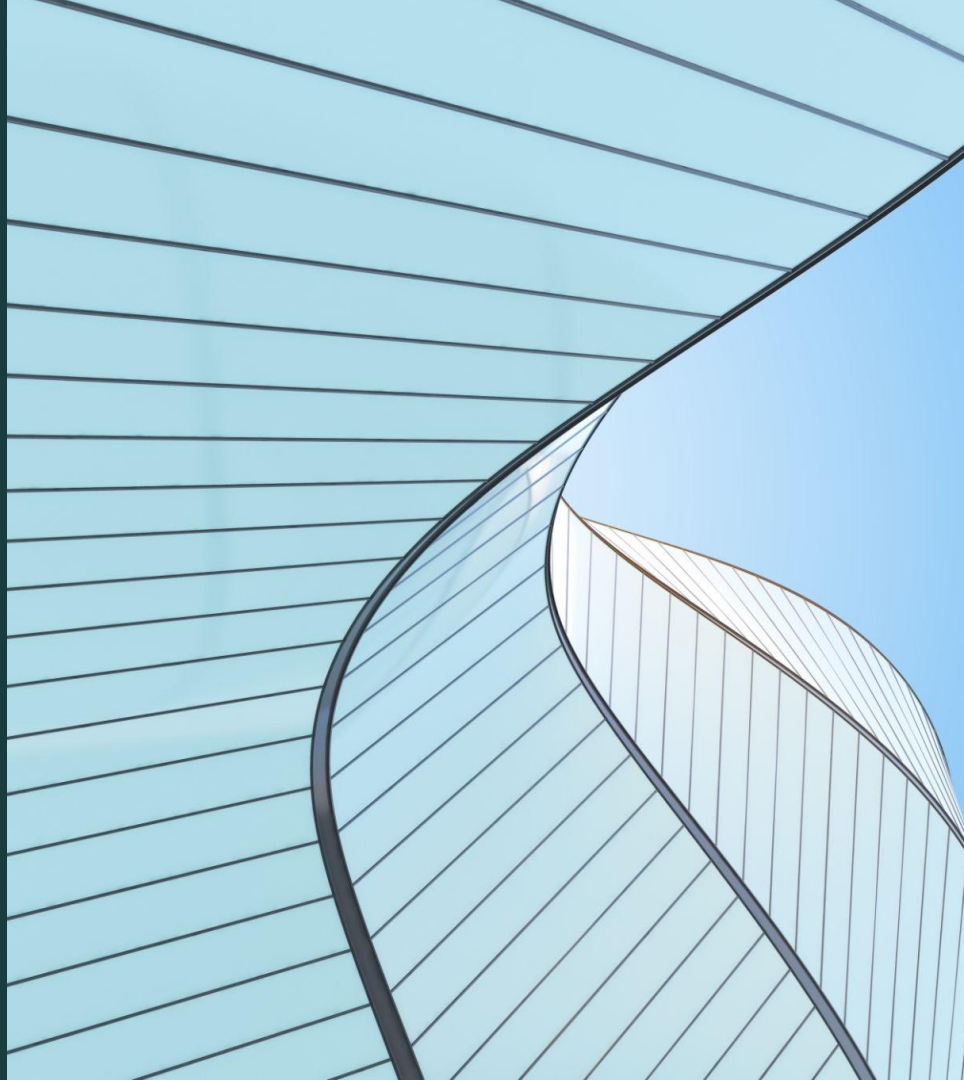


<u>Goal:</u>
Analyze whether graduates who move to high-cost states experience real income gains or whether higher nominal wages are offset by higher living expenses

Outline

I. Datasets Used

II. Data Analysis/Research

III. Interesting Questions

IV. Conclusion

Datasets Used

Adult Census Income - Survey used to track individual age, education, occupation, hours worked, income amount, and more.

US Cost of Living (Kaggle) - County-level cost of living broken down into housing, food, transportation, healthcare, childcare, and taxes.

US jobs on Monster.com (Kaggle) - Data scraped from Monster.com regarding job titles, location, salary, and sector (i.e. private/public).

Public Use Microdata Sample (PUMS) - This is the American Community Survey Public Use Microdata, which holds information such as income, state, education, occupation, and migration status.

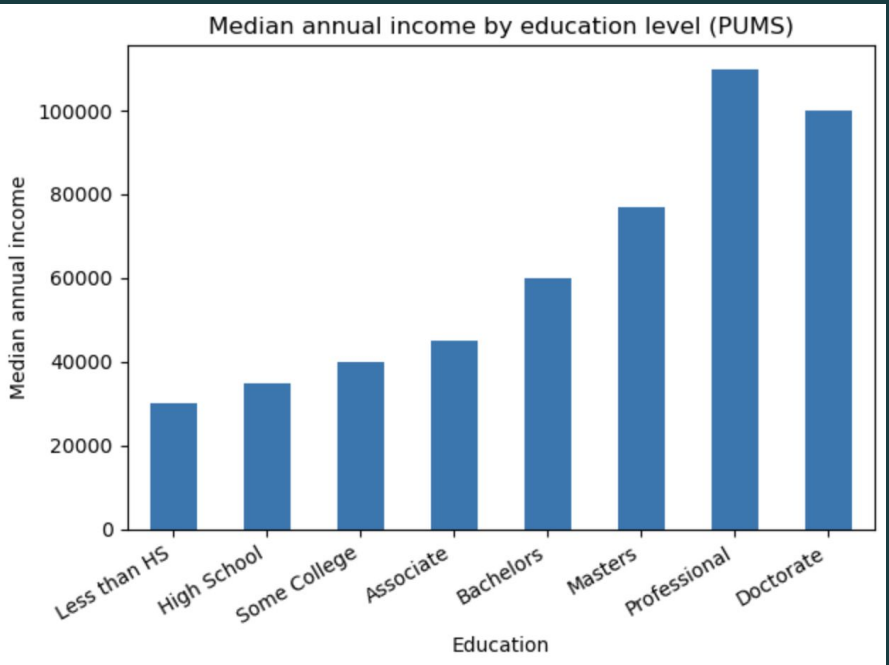
IRS SOI Migration Data - State-to-state migration flows for tax year 2021→2022, used to quantify which states people are actually moving to and from, and at what income levels.

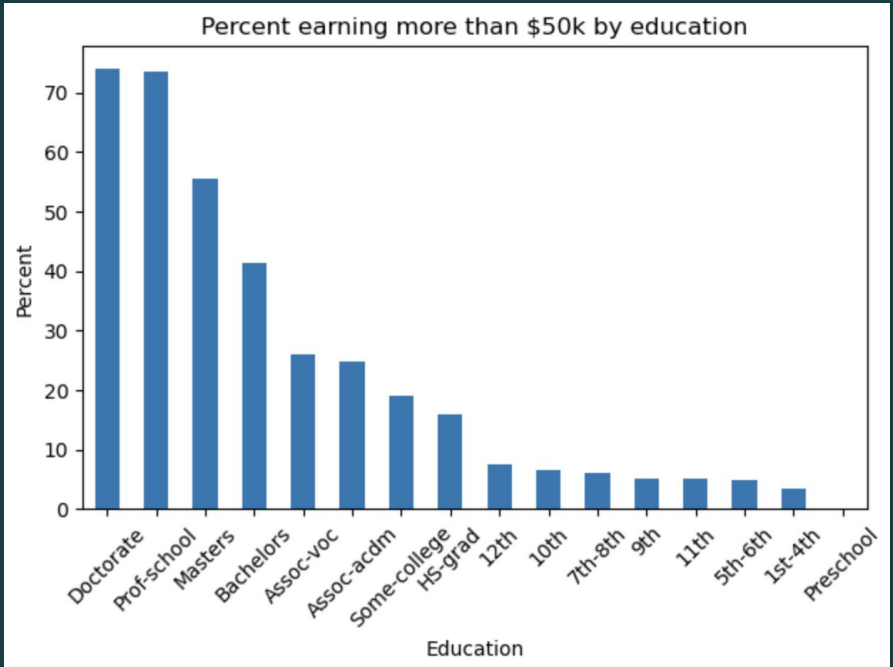
BLS Wage Files - State-to-state migration flows for tax year 2021→2022, used to quantify which states people are actually moving to and from, and at what income levels.

- wm_data_1_AllData.csv:
  - Contains the actual wage amounts for occupations
- wm_area.csv
  - Used to get state/metro info for each wage record
- wm_occupation.csv
  - Connect series ID to occupation
- wm_subcell.csv
  - Used to filter down to summary-level records only, to avoid double-counting breakdowns like union vs non-union workers

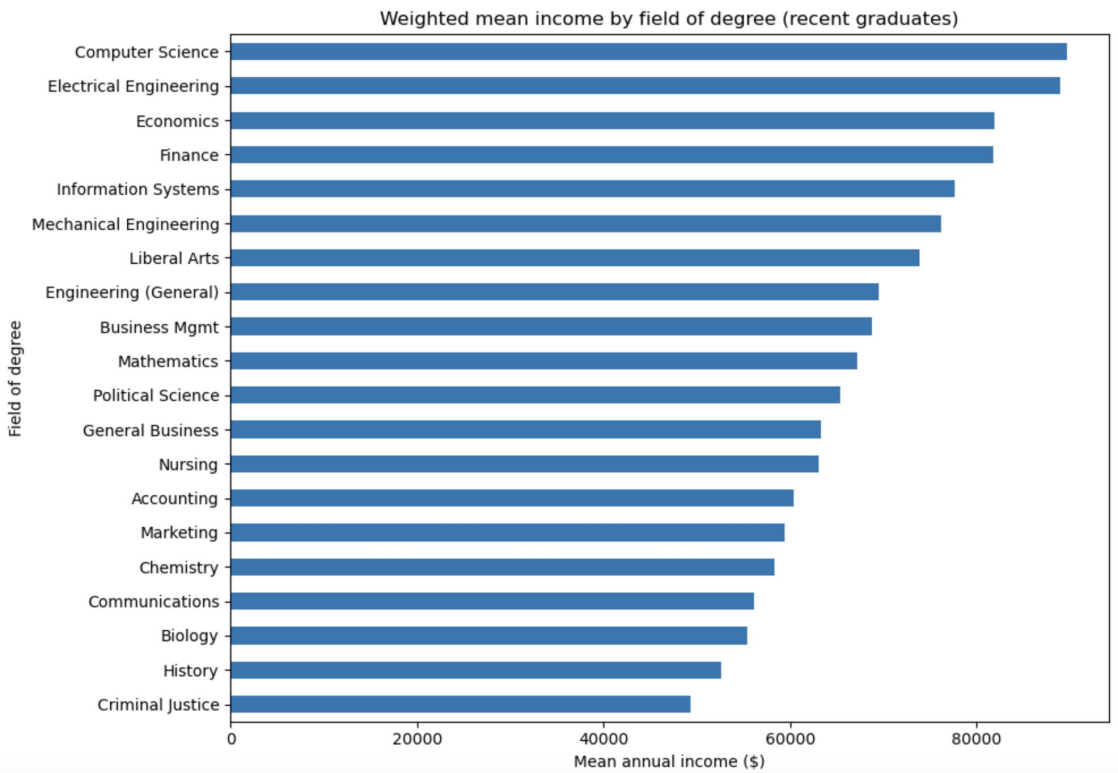
Education and Income

- Common assumption: the higher the education level, the greater the financial return
- The Adult Census confirms this, using its >$50k threshold, those with Doctorates or Professional degrees (e.g. Medicine, Law, Dentistry) show the widest earning margins
- The PUMS dataset reinforces this hypothesis by supplementing real dollar amounts rather than just a threshold
  - Even with exact salaries, we still see a slight edge towards Professional degrees in raw income power





- Common assumption: certain fields yield a higher financial return after graduation
- PUMS data confirms this using weighted mean income for recent graduates
- Computer Science and Electrical Engineering lead at roughly $90k, nearly double Criminal Justice at $49k
- Economics and Finance closely follow at $82k, outranking most engineering disciplines
- Key takeaway: field of study can matter just as much as degree level

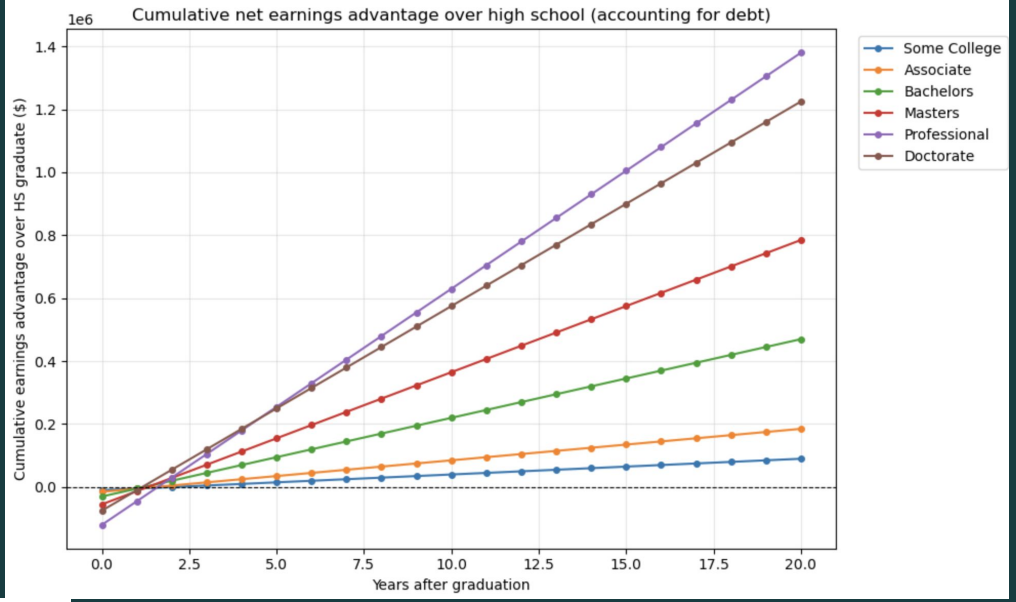


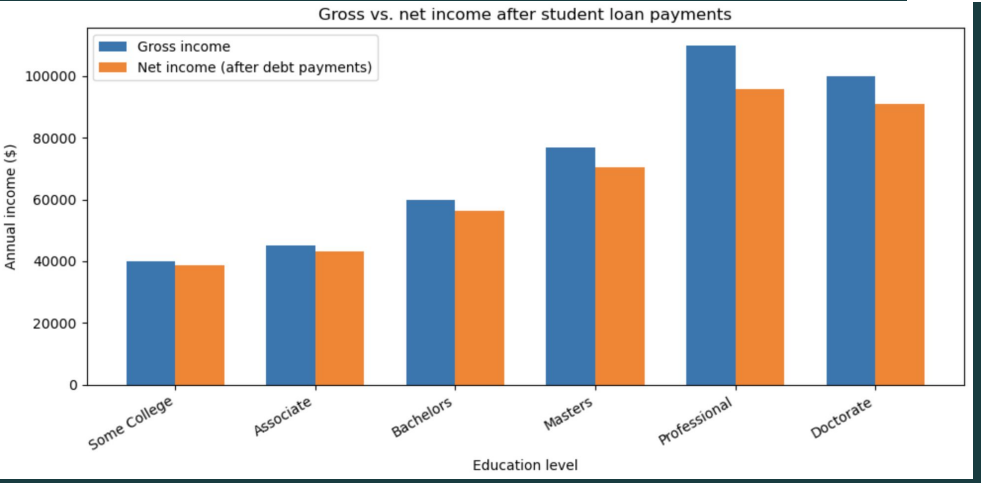
What you Study Matters

Does the Debt Pay Off?

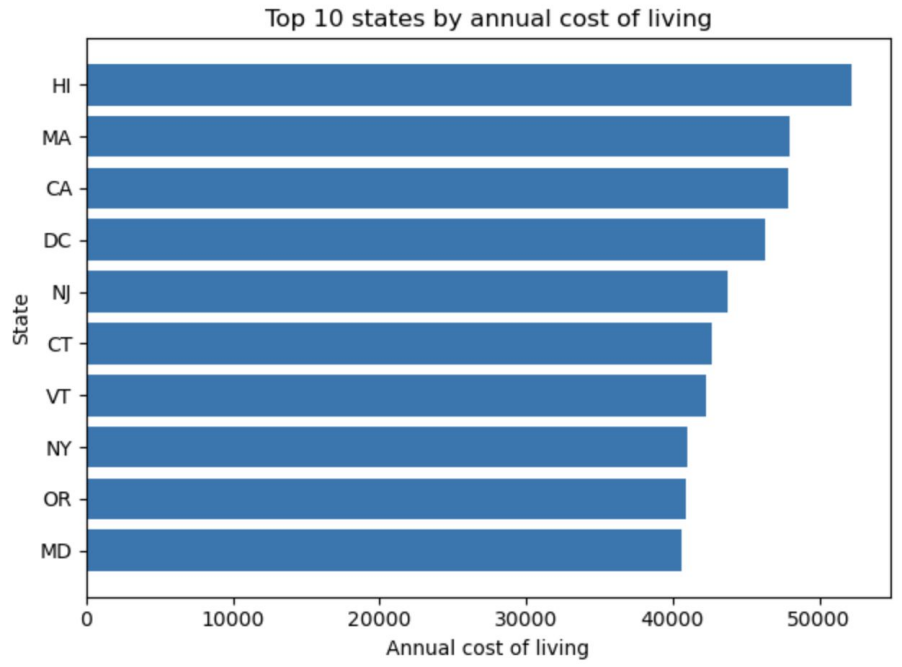
Many students also end degree study with Student Debt:

- We compared the gross income to net income at certain degree levels
- Notice that more education results in a great difference in gross income vs net income

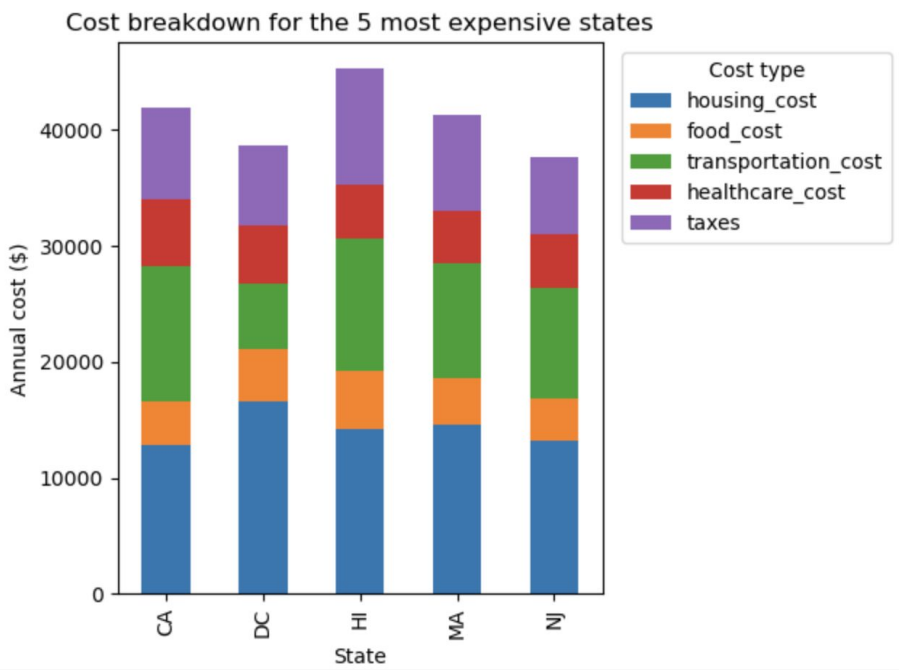




- All degree levels break even within 3-5 years, validating the upfront cost of higher education
- After 20 years, Professional degrees lead at ~$1.4M cumulative advantage over a high school graduate



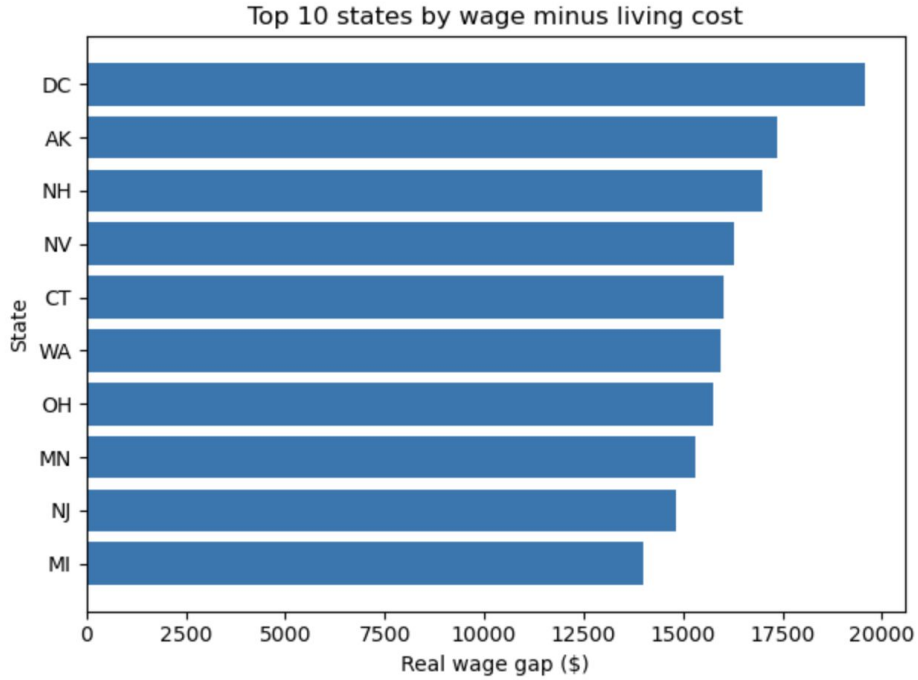
- Cost of living varies significantly by state and directly affects real purchasing power
- The three most expensive states are Hawaii, Massachusetts, and California
- Housing and transportation are the largest cost drivers, with taxes as a significant secondary factor

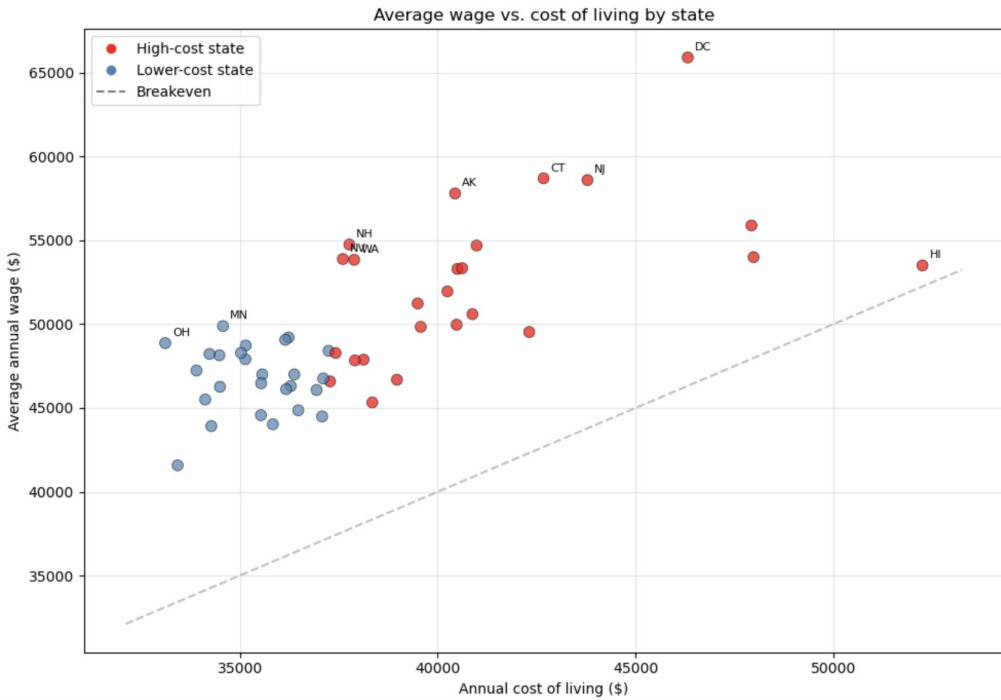


The Cost of Living Problem

Real Wage Gap

DC leads the real wage gap at nearly $20k, ahead of AK and NH, none traditionally known as the highest paying states

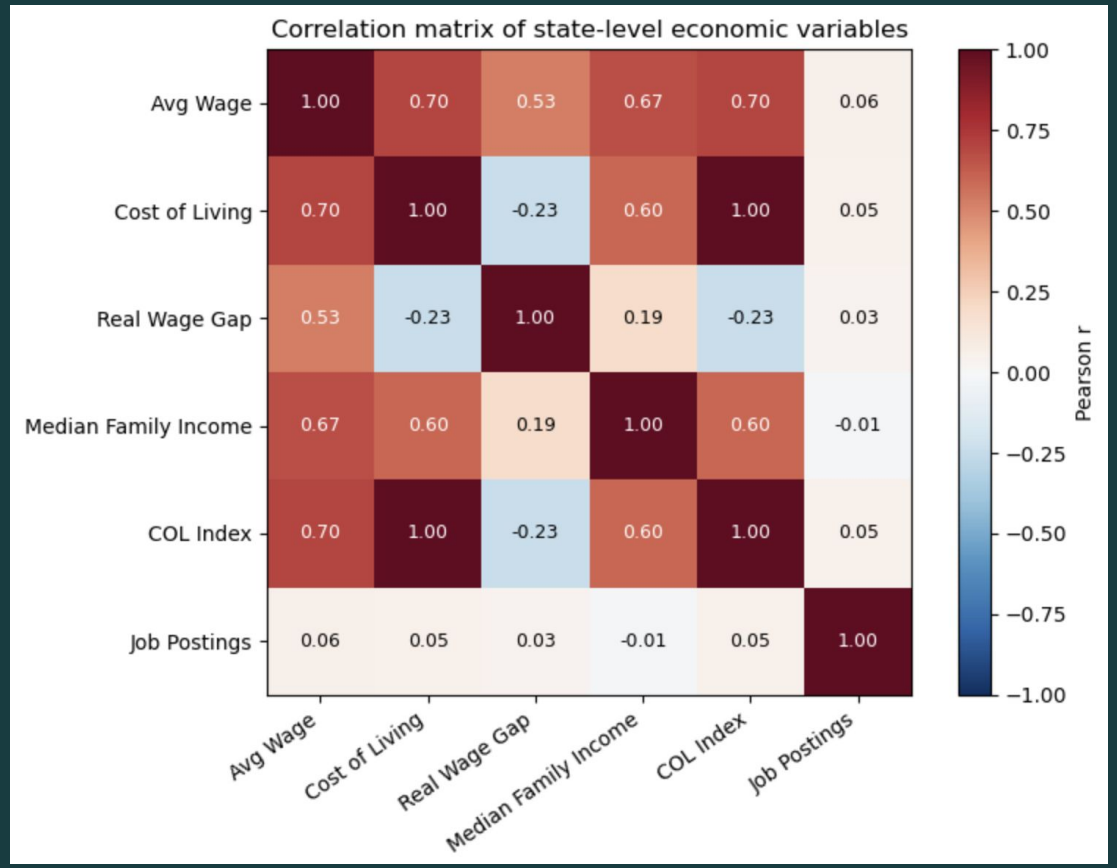




- Every state sits above the breakeven line, but high-cost states like Hawaii barely clear it despite high nominal wages
- Lower-cost states consistently show a larger gap from the breakeven line than expensive coastal states
- Key takeaway: highest nominal salary does not guarantee the best financial outcome

Putting it Together

- Avg Wage and Cost of Living are strongly correlated at r = 0.70, states that pay more also cost more
- Real Wage Gap vs Cost of Living shows r = −0.23, higher cost states actually tend to have a smaller real wage gap
- Job Postings show near zero correlation with all other variables, meaning volume of opportunities has no relationship to actual wages or cost of living
- Key takeaway: high wages and high cost of living cancel each other out, reinforcing the real wage gap as the most meaningful metric



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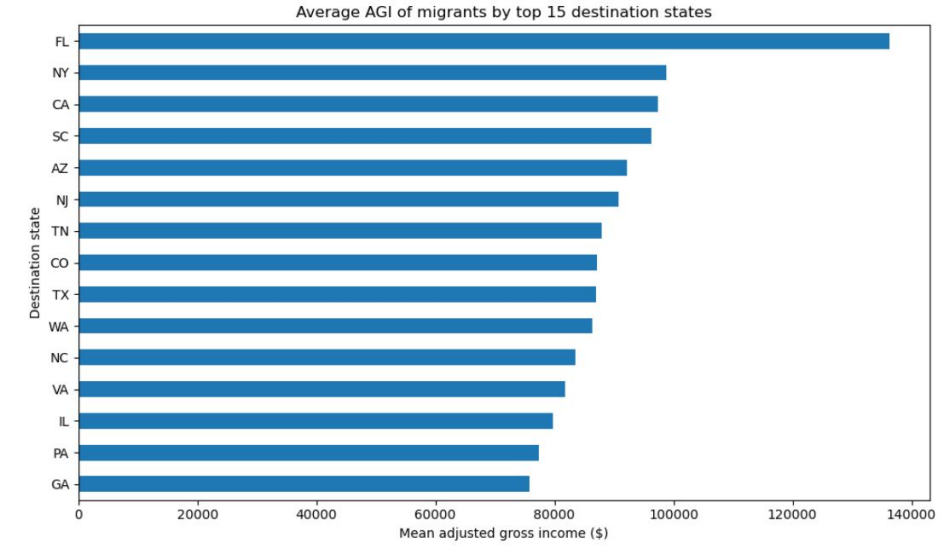
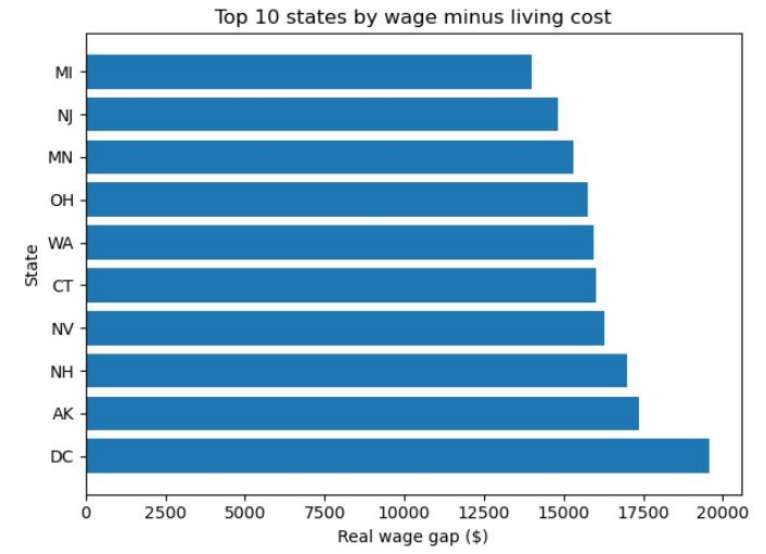


Nominal Income (The Illusion)



Real Purchasing Power (The Reality)

- **The "Paper Wealth" Paradox:** High Adjusted Growth Income (AGI) is maybe driven by elite industry clusters (Tech/Finance) or skewed by high-net-worth residents and no state income tax.

- **Analysis:** Gross income (AGI) measures wealth accumulation at the top, but the Real Wage Gap measures the survivability and savings potential for the average professional.

Where are people typically migrating to?

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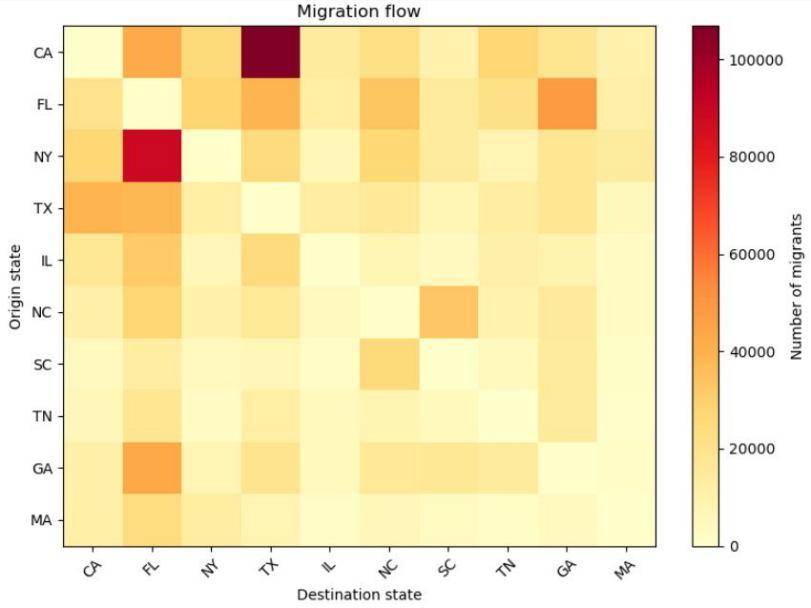

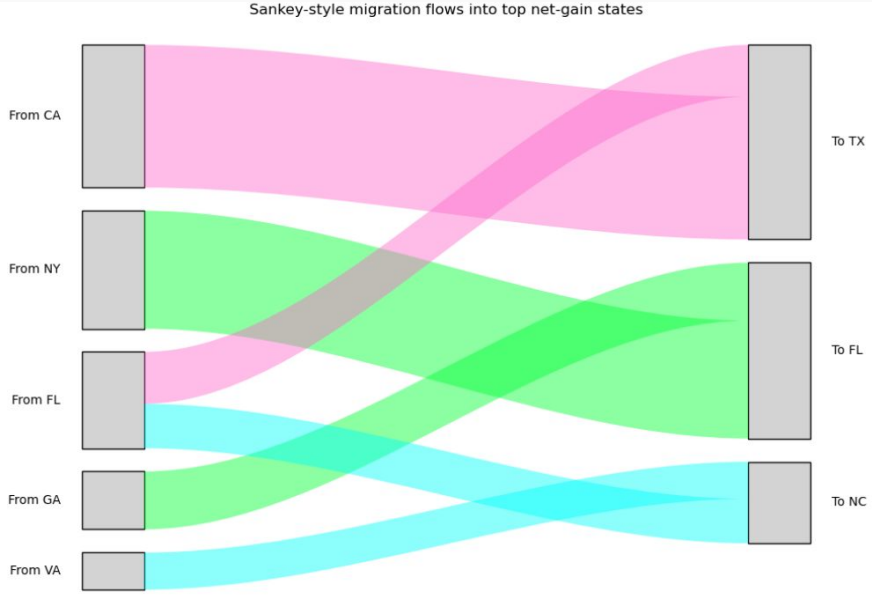

- **The "Sun Belt" Trend:** There is a massive shift from the Northeast and West Coast toward the Sun Belt (FL, TX, AZ), driven by lower taxes and more affordable housing.

- **Targeted Corridors:** Our Sankey diagram reveals specific "migration corridors," such as NY→FL and CA→TX.

- **Economic Rationality:** People aren't just moving randomly; they are following the Real Wage Gap. Even if nominal salaries are lower in TX than CA, the increase in disposable income makes the move financially rational for debt-heavy graduates.

Does level of education and age affect income?

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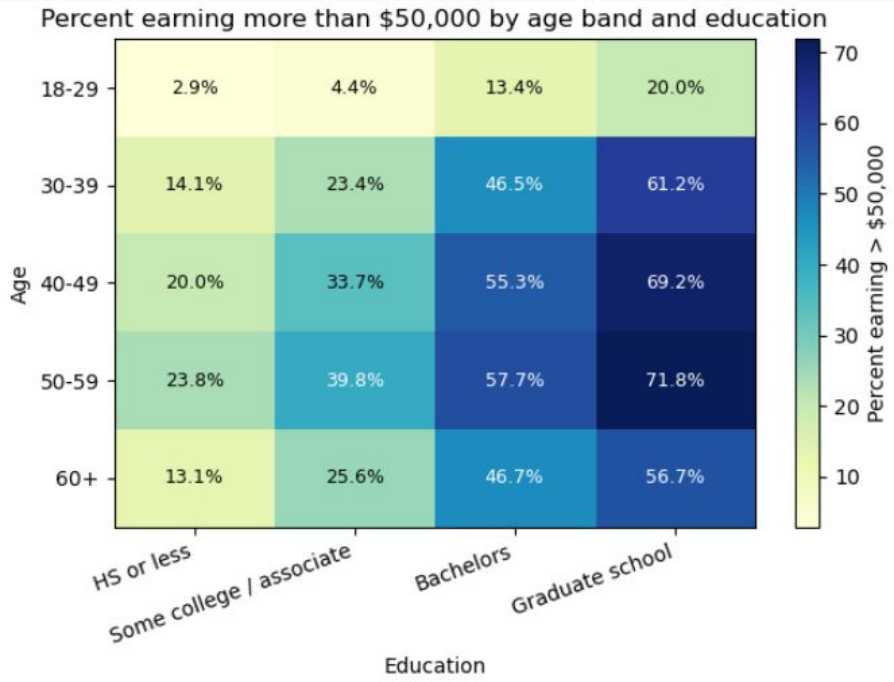
- **The "Graduate Premium":** The "Graduate school" category (combining Masters, Professional, and Doctorate) shows the highest probability of earning > $50k, exceeding 50% during peak career years.

- **Growth Potential:** While all groups see gains with age, the Bachelors and Graduate groups experience the most aggressive growth between ages 30 and 49.

- **Stagnation Risk:** Individuals in the "HS or less" category face a significant income ceiling, with fewer than 1 in 4 reaching the 50k threshold regardless of experience.

- **Strategic Takeaway:** The data justifies the initial debt of a Graduate degree as it secures a high-income trajectory that persists into later career stages.



Goal:

Analyze whether graduates who move to high-cost states experience real income gains or whether higher nominal wages are offset by higher living expenses

The Verdict: Higher nominal wages are often **NOT** enough to offset living expenses in many "Tier 1" states

Key Evidence:

- **High Correlation (r ≈ 0.88):** Wages and Cost of Living are tightly coupled. You aren't just getting paid more; you are being compensated for the "cost of existence"
- **The "Efficiency" Leaders:** DC, AK, and NH are the rare exceptions where the wage growth significantly outpaces the cost curve
- **The "High-Cost Trap":** In states like California and Hawaii, graduates experience a Real Income Loss compared to lower-cost, high-gap states
- **Migration Insight:** Our data shows that while people move toward "opportunity," many are essentially "running on a treadmill"—earning more but saving less

Final Recommendation: Graduates should prioritize "Real Wage Gap" over "Nominal Salary" when evaluating job offers to maximize long-term wealth

Thank you!